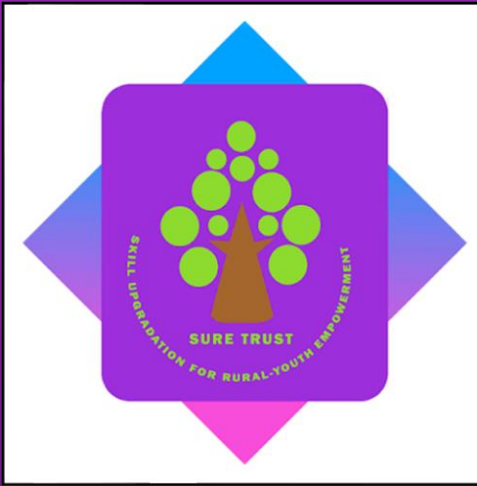




# **SURE ProEd**

**A UNIT OF SURE TRUST**

**SKILL UPGRADATION FOR RURAL-YOUTH EMPOWERMENT – TRUST**

## **INTERNSHIP PROJECT ON:- SOLAR PANEL CLEANING SYSTEM**

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# INTRODUCTION OF THE MODEL

- Solar panels need to be clean to work efficiently and produce maximum electricity.
- Dust, dirt, bird droppings, and leaves reduce the power output of solar panels.
- Manual cleaning is time-consuming, costly, and not practical for large systems.
- An automatic cleaning system helps keep panels clean without human effort.
- This project uses a camera sensor to detect dirt on the solar panel.
- When the camera detects dirt, the system automatically starts cleaning (e.g., with a wiper or brush).
- It saves time, reduces labor cost, and improves solar panel performance.
- The system is smart, efficient, and suitable for homes, buildings, and solar farms.

# LITERATURE REVIEW

| Sl. No. | Author(S)                  | Work Carried Out                                                                                           |
|---------|----------------------------|------------------------------------------------------------------------------------------------------------|
| 1       | Manju et al. [ 2018]       | Automatic Solar Panel Cleaning System                                                                      |
| 2       | Minh Phu et al. [2024]     | Research, design and manufacture a prototype of solar panel cleaning system with semi automatic controller |
| 3       | Patil et al. [2023]        | Design and Analysis of Automated Solar Panel Cleaning System                                               |
| 4       | Zatsarinnaya et al. [2020] | Automated system for cleaning solar panels based on a linear actuator                                      |
| 5       | Kar et al. [2020]          | Design and development of waterless solar panel cleaning system                                            |

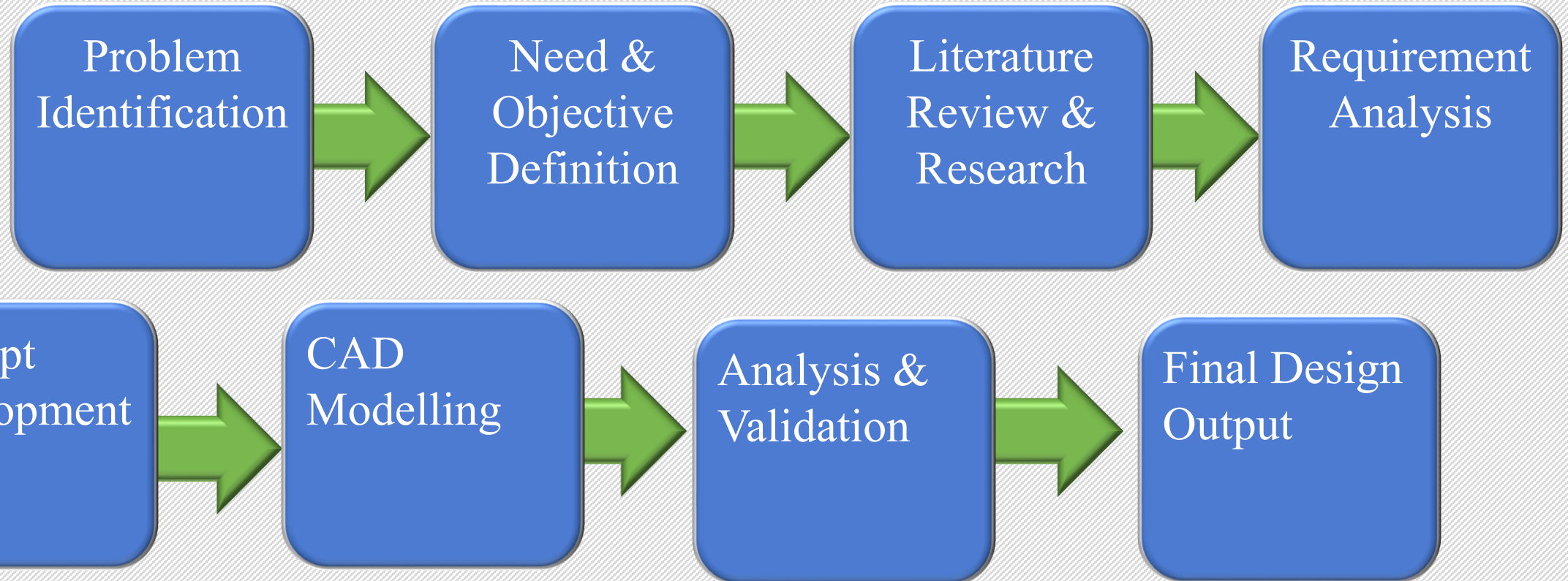
# PROBLEM STATEMENT

- Solar panels are widely used to generate clean and renewable electricity. However, dust, dirt, bird droppings, and other particles accumulate on the surface of solar panels over time. This reduces the amount of sunlight reaching the photovoltaic cells and consequently decreases the efficiency and power output of the solar panel.
- Manual cleaning requires significant time, labour, and water, especially in large solar installations. Therefore, there is a need for an efficient, economical, and automated solar panel cleaning system that can regularly remove dust and dirt from the panel surface with minimum human intervention. The proposed system aims to improve solar panel performance, reduce maintenance effort, and maintain maximum power generation through an effective cleaning mechanism.

# OBJECTIVES

- To design and develop an automated solar panel cleaning system to remove dust, dirt, and other particles from the panel surface.
- To improve the efficiency of solar panels by maintaining a clean surface for maximum absorption of sunlight.
- To reduce manual labour and maintenance time required for cleaning solar panels.
- To develop a cost-effective cleaning mechanism suitable for domestic as well as large-scale solar installations.
- To minimize water consumption during the cleaning process.
- To ensure safe and effective cleaning without causing damage to the solar panel surface.
- To improve the overall power generation and performance of the solar energy system.
- To create a reliable system with minimum human intervention for regular solar panel maintenance.

# METHODOLOGY



# 3D DESIGN

MOTOR

BRUSH

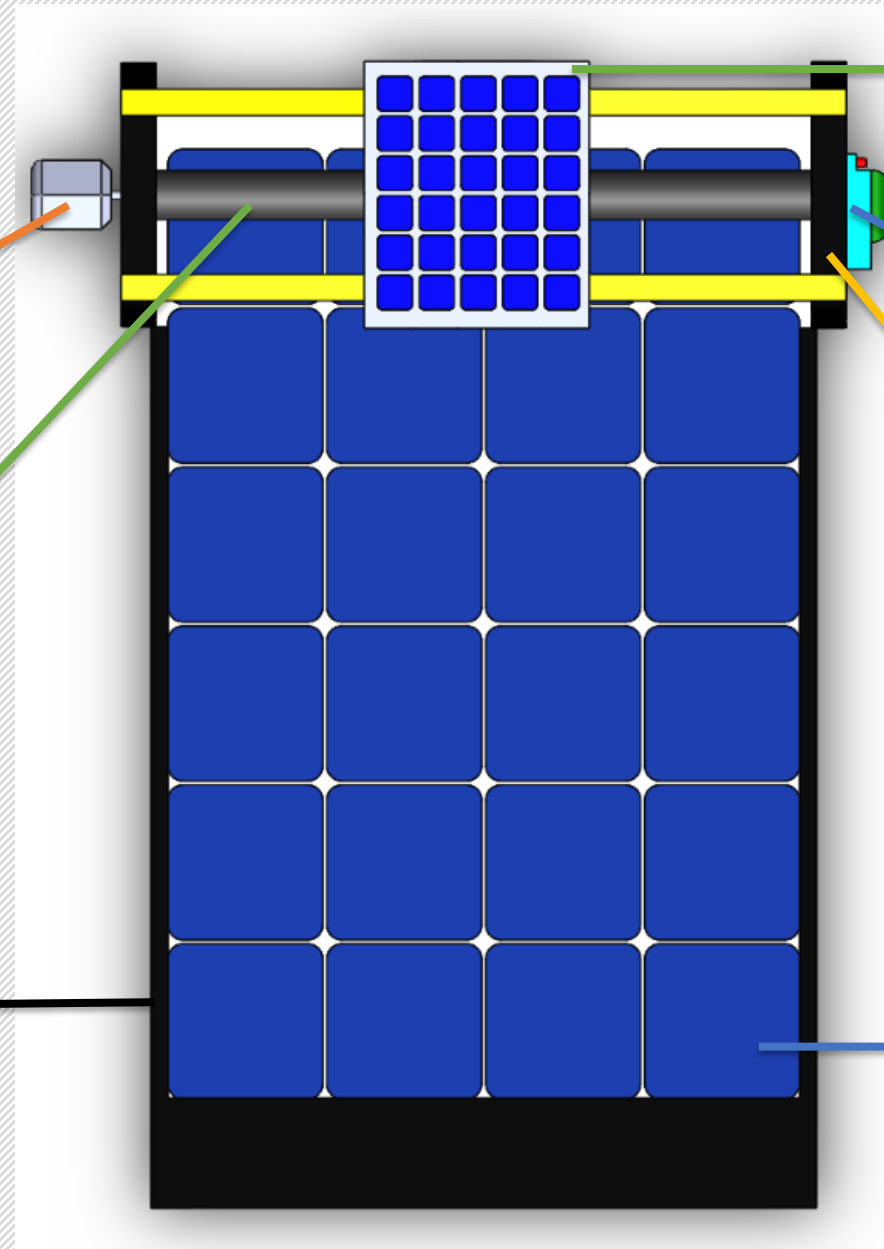
CLEANING PLATFROM  
FOR SOLAR PANEL

SOLAR MODEL FOR  
POWER SUPPLY

RELAY MODEL

ROTATING SYSTEM OF  
CLEANING SYSTEM

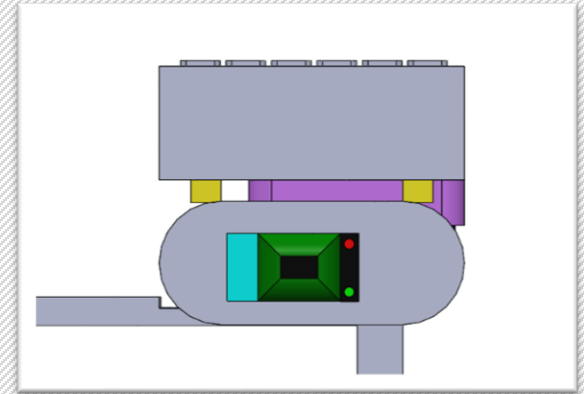
SOLAR PANEL



# FUNCTIONS OF COMPONENTS

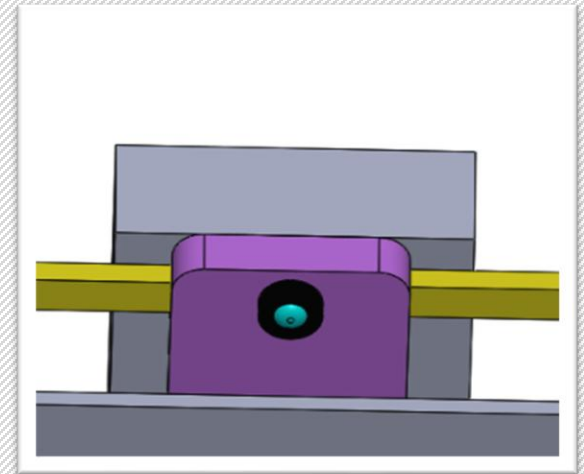
## RELAY MODEL

- Turn electrical devices ON/OFF
- Control high-power devices using a low-power signal
- Provide electrical isolation between control and power circuits
- Work automatically with sensors and programmed systems
- Control devices such as motors, pumps, lights, and other electrical equipment



## CAMERA SENSOR

- Capture photos
- Stream live video over Wi-Fi
- Detect/monitor objects (with programming)
- Send images and data through Wi-Fi
- Work with sensors and automation systems

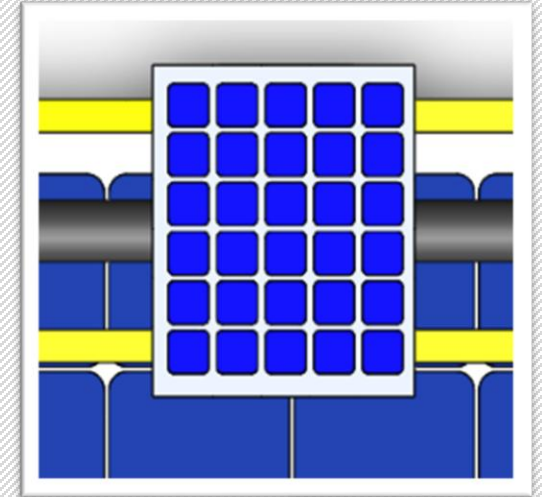




# FUNCTIONS OF COMPONENTS

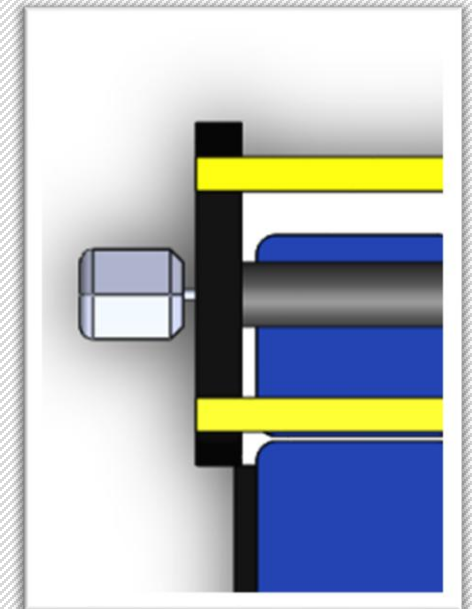
## SOLAR MODEL FOR POWER SUPPLY

- Convert sunlight into electrical energy
- Supply electrical power to the system
- Charge the battery for energy storage
- Provide power to electronic components and motors
- Enable the system to operate using renewable solar energy

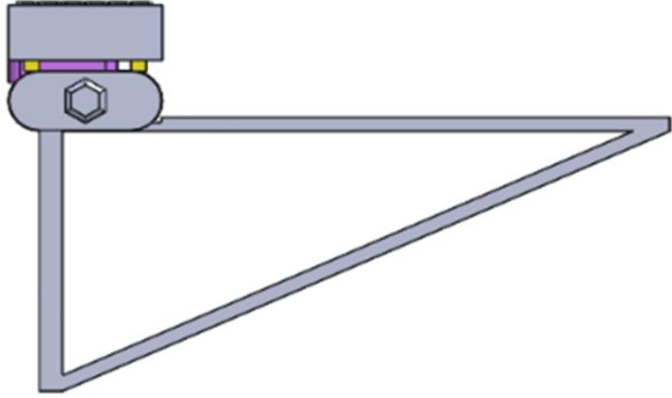


## MOTOR

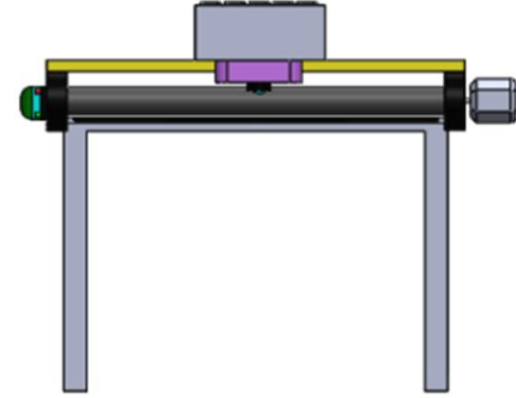
- Convert electrical energy into mechanical motion
- Provide rotational or linear movement for the cleaning mechanism
- Drive the brush or cleaning mechanism across the solar panel
- Work automatically based on the control system signals
- Enable efficient movement and operation of the solar panel cleaning system



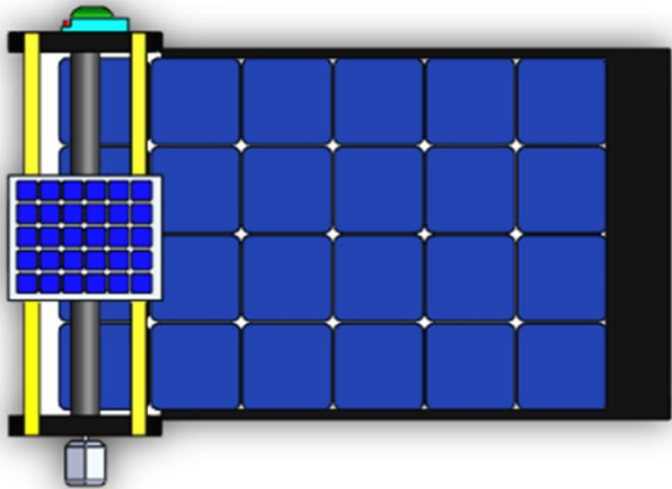
# ALL VIEWS OF THE MODEL



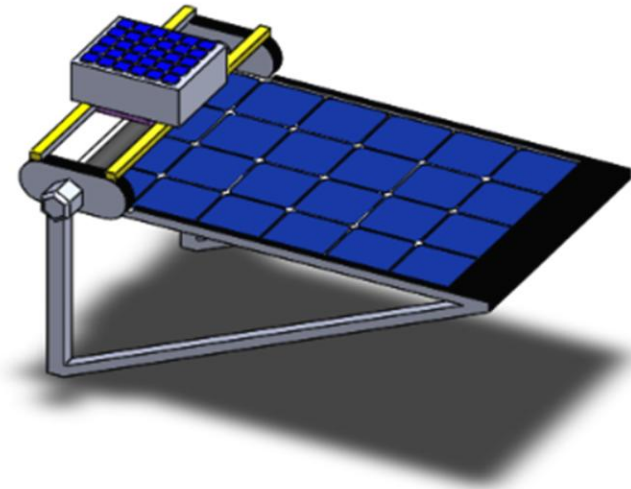
\*Front



\*Left



\*Top



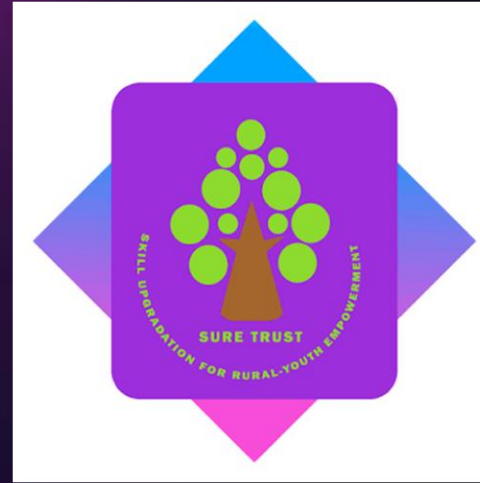
\*Trimetric

# FUTURE WORK

The future work of the solar panel cleaning system can focus on improving its automation, efficiency, and reliability. Dust and environmental sensors can be integrated to automatically detect dirt accumulation and initiate the cleaning process when required. IoT-based monitoring can allow users to control and monitor the system remotely through a mobile application. The system can also be powered using solar energy generated by the panel itself, making it more energy-efficient and sustainable. Future developments may include waterless cleaning mechanisms, weather-based control, AI-based cleaning schedules, and improved robotic movement for different sizes and orientations of solar panels. The system can also be further developed for large-scale solar power plants to clean multiple panels automatically while reducing maintenance cost and improving overall power generation efficiency.

# CONCLUSION

The solar panel cleaning system is designed to provide an effective and efficient solution for removing dust, dirt, and other particles accumulated on the surface of solar panels. Regular cleaning helps maintain maximum sunlight absorption and improves the overall power generation efficiency of the solar panel. The proposed system reduces manual effort, saves time, and provides a safer and more convenient cleaning method. By using an automated mechanism, the system can perform cleaning operations efficiently and consistently. Overall, the solar panel cleaning system is a practical, cost-effective, and sustainable solution that can help improve the performance and lifespan of solar power systems.



THANK YOU